An Investigation into Dashboard Usability

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The present paper aims to investigate consumer perceptions, feelings, and usability rankings of vehicle dashboard controls to establish users' pain points in the face of shifting industry trends that focus on implementing more screen interfaces. To this end, a mixed-methods survey was distributed, receiving a completed n = 67 results. Users' demographic information was collected alongside their vehicle types. For this study, vehicles were broadly categorized into three main groups: analog, digital, and hybrid based upon presentation characteristics and functionality. Likewise, each user's vehicles were accordingly ranked using a modified System Usability Scale (SUS). It was found that digital dashboards ranked poorly among most users, and the traditionally lauded analog controls faced similar scrutiny among many consumers; the mixed-interface hybrid models were the most positively received. Using these results, further research must be conducted to quantify user efficiency ratings, optimal control placements, as well as creating a standard for vehicle interfaces.

INTRODUCTION

Automotive dashboards have shifted from analog simplicity to hybrid systems and fully digital interfaces, fundamentally altering how drivers interact with vehicles. Analog dashboards, built on physical buttons and switches, offer tactile feedback and ease of use, relying on muscle memory for quick access. Hybrid dashboards combine analog controls with digital features, blending familiarity with modern functionality. Fully digital dashboards focus on features like touchscreen interfaces and dynamic layouts but often create challenges related to usability and cognitive demands.

Research in Human Factors and Ergonomics (HFE) shows how user-centered design improves automotive dashboards. Physical ergonomics, such as the placement and accessibility of controls, reduces the driver's cognitive workload. Steering wheel controls, for example, are widely appreciated for their convenience, while controls near the gear stick often need to be higher in usability. Cognitive ergonomics looks at how drivers handle information, with studies showing that cluttered displays and poorly grouped elements overwhelm drivers and slow their reaction times. Logical control layouts and functional groupings help reduce drivers' time looking away from the road (Gibson, Butterfield, & Marzano, 2016).

Digital systems, while technologically advanced, present safety concerns about driver attention. A study by AAA found that tasks like programming navigation on digital dashboards can take up to 40 seconds, a delay comparable to texting while driving. Similarly, in-vehicle information systems (IVIS) and navigation systems, although useful, are often criticized for being overly complex and distracting. These findings illustrate the challenge of integrating new technology into dashboards without reducing usability or safety (Gibson et al., 2016). The rise of automated vehicles introduces new challenges. Many drivers remain skeptical about the safety and reliability of automation, citing fears of losing control. However, automation offers benefits such as reducing stress and increasing mobility. Tailored dashboard configurations that address public hesitation could help build trust and encourage adoption (Gibson et al., 2016).

Key questions for this research: How do drivers perceive the usability of analog, hybrid, and digital dashboards? How does prior experience with specific systems shape preferences? How do emerging safety and usability trends align with the industry's shift toward automation and fully digital interfaces? By exploring these questions, this research looks to influence future dashboard designs, balancing technology with safety, usability, and driver satisfaction.

METHOD

Participants

Participants were recruited from San Jose State University students as well as friends, family, and coworkers of the researchers in California and Nevada.

For this survey, participants needed to meet the eligibility requirements of being able to consent, being 18 or older, and having a valid government-issued driver's license to participate. By the cutoff date, the survey recruited 77 participants, of which only 67 had completed the survey.

Participants were roughly evenly split in gender, with 50.7% of participants identifying as male, 47.8% identifying as female, and 1.5% preferring not to disclose that information. Participants' age demographics were grouped within different age ranges, of which 38.8% were between the ages of 18 to 24,

14.9% were 25-34, 14.9% were between 35-44, 26.9% were 45-54, and 4.5% were 55 or older.

As it pertained to driving experience, 11.9% had only driven for 0-2 years post license, 20.9% have driven for 3-5 years, 19.4% for 5-10 years, and 47.8% of which have driven for over 10 years. Participants also identified differences in ownership experience, with participants ranging in number of cars owned. In our participant group 16.4% had only owned a single car, 40.3% have owned 2 or 3 cars, 16.4% owning 4-5 cars, and 26.9% of participants owning over 5 cars to date.

Procedure

Using Qualtrics, the researchers created a 27-question survey to gather both demographic and vehicle usability information, with an estimated completion time ranging from 10 to 20 minutes. Questions consisted of multiple choice as well as 5-point Likert-scale questions to assess usability, based upon a modified System Usability Scale (SUS).

If applicable, participants had to enter the year, make, and model of their currently owned and most operated vehicle (to the best of their knowledge), and that information was piped into the SUS questions to enhance recollection and feelings towards their preferences and usability ratings. For instance, if the participant owned a 2007 Honda Civic, they would enter each portion into separate questionnaire blocks to sort through later. In the event that a participant did not currently own a vehicle, the survey was shifted to reflect upon the most familiar vehicle they have driven for work or license-obtaining purposes. The questions were reduced to 7 SUS questions for participant retention purposes.

Questions primarily targeted location and accessibility of volume controls and the like that drivers may want to use, and asked about how comfortable or confident they felt doing so in the context of driving, on a scale of “Strongly Disagree” to “Strongly Agree.” Participants were asked to rate their agreement with statements like: “I feel confident in my ability to operate this vehicle,” or “I have found myself unable to locate necessary buttons or screens while driving.” In addition to the SUS questions, participants were given 3 targeted questions directly asking how they felt about each modality of dashboard control in the form of “I prefer -style” questions on the same Likert scale.

The participants were also given an optional open-ended question in which they could voice certain opinions related to the topic of car dashboards or controls that they had felt were important to consider or were skipped over during this survey.

The survey was open from the period of November 12th, 2024 to November 24th, 2024, and distributed through San Jose State University’s SAMMY app, the Human Factors and Ergonomics Society (San Jose State University Chapter) Discord server, as well as through email and QR Code distribution among friends, family, and coworkers.

Data Analysis

By the cutoff date, participants exported the results into SPSS and into Google Sheets for analysis. Participants’ vehicle information of year make and model was separately exported and manually categorized by the researchers to fit into the categories of “analog,” “hybrid,” or “digital” dashboard modalities, and re-matched to their data sets. Within SPSS, the researchers created an overall usability score using information from the SUS to create a general score out of 35, converted to a score out of 100, and were then ranked among each other.

statistical tests were conducted at an alpha level of .05. For the first hypothesis, an independent t-test was conducted. Given the amount of participants gathered in each category, digital and hybrid users were combined into one group, and compared against the analog ones. This was also important when factoring in the presence of a capacitive screen. Once grouped, participant’s usability scores were compared against one another. For the second hypothesis, a Spearman’s correlation was conducted from the participant’s ranking of the targeted questions, and their currently owned vehicle category to see if there was any correlation.

RESULTS

Trends in Vehicle Types

Beginning with the participant’s car ownership type seen in Figure 1., the participant’s vehicle type tends to align with the current industry trends of involving some sort of touch screen capability within most vehicles. 37.3% of participants were found to have an analog vehicle, 53.7% had a hybrid type, and 6% had a digital-only vehicle. Of our participants, 2% had no vehicle and thus were categorized as not applicable

Usability Rankings

In calculating usability scores, as seen in Figure 2., analog groups scored an average score of 82.17 (SD = 9.18), hybrid users scored an average of 86.90 (SD = 6.12), and digital users scored an average of 90.00 (SD = 3.69).

When split across gender, the average usability score for males was 84.66 (SD = 3.30) whereas the average score for females was 86.54 (SD = 1.97).

Hypothesis 1 – Usability scores

For our first prediction, we found that analog scores were actually lower than their hybrid and digital counterparts, $t(63) = -2.145$, $p = 0.018$. Analog scores are actually moderately lower in score in comparison to digital-like displays by .547 std deviations.

Hypothesis 2 – Association of ownership and preferences

For our second prediction, it appears that ownership types do have slight correlations with ranking, however, there was only one significant correlation; Participants ranked analog-only scores lower than their hybrid and digital-only counterparts $r(67) = -0.294$, $p = 0.017$

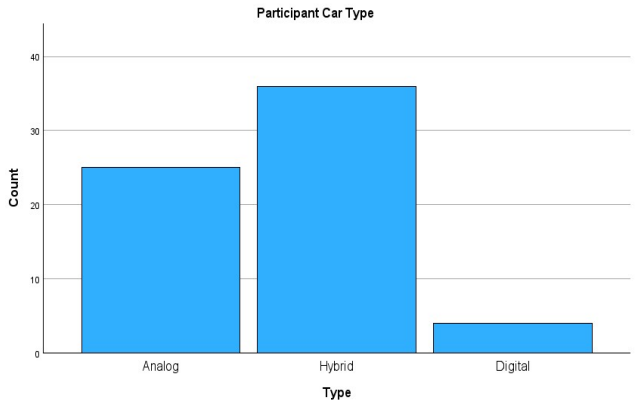


Figure 1. Participant car ownership types

DISCUSSION

This study explored how drivers perceive and interact with analog, hybrid, and digital dashboards, identifying usability trends and challenges that influence design decisions.

How Do Drivers Perceive the Usability of Different Dashboard Styles?

Participants shared varied opinions about dashboard types. Analog dashboards, used by 37.3% of participants, were rated lowest in usability (82.17), valued for their straightforward design but noted for limited functionality. Hybrid dashboards, favored by 53.7%, earned the highest usability score (86.90), balancing traditional tactile controls with modern digital features. Digital dashboards, although used by only 6% of participants, scored the highest (90.00) but were sometimes described as overly complex and time-consuming to navigate.

How Does Prior Experience Affect Preferences?

Participants familiar with hybrid or digital systems rated them higher in usability, while analog drivers valued tactile feedback but rated these systems lower overall compared to their hybrid and digital counterparts. This shows the need for designs that balance familiarity and innovation, particularly for drivers transitioning from analog to more advanced hybrid and digital systems.

What Safety and Usability Trends Appear?

Digital dashboards pose notable safety risks, often requiring significant driver attention to operate. This aligns with AAA findings that programming navigation tasks can take up to 40 seconds, a time comparable to texting while driving. Hybrid systems mitigated some of these risks by retaining tactile controls for essential functions.

Challenges Identified

Several challenges emerged during this study:

- Safety Concerns: Digital dashboards increase the risk of distraction due to their complexity, making reducing cognitive load a critical goal for future designs.
- Sample Limitations: The sample size of 67 participants may not represent the full diversity of driver experiences. Over half (53%) of respondents drove vehicles from two Japanese automakers, Toyota and Honda, which may have influenced preferences due to brand-specific designs.
- Balancing Features with Usability: While digital features are desirable, they must be implemented without overwhelming the driver or increasing distraction.

Implications for Automotive Design

The findings highlight several opportunities to improve dashboard usability while addressing user concerns:

- Streamline Digital Interfaces: Simplifying layouts and reducing unnecessary visual elements can make critical functions easier to find and faster to use. For example, grouping frequently used features like navigation and media controls in logical clusters can help reduce the time drivers spend interacting with the dashboard and navigating menus.
- Enhance Hybrid Systems: Combining tactile controls with intuitive digital features provides a user-friendly experience for a diverse group of drivers. Retaining physical knobs and buttons for key functions like climate control and audio volume can reduce the learning curve for drivers transitioning from analog systems.
- Offer Customization Options: Providing the ability to adjust dashboard settings allows drivers to personalize their experience based on their needs and preferences. This not only improves satisfaction but also helps drivers quickly adapt to the system. By addressing these areas, automotive designers can create dashboards that meet both practical and aesthetic needs while enhancing driver safety and usability.

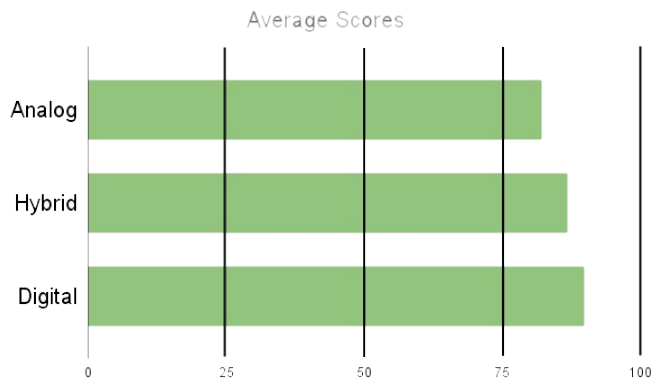


Figure 2. Participant car usability scores by type

Conclusion and Recommendations

This study highlights the importance of balancing usability, safety, and innovation in automotive dashboard design. Hybrid systems, which achieved the highest usability scores (86.90), demonstrate the value of combining tactile feedback with digital functionality. Digital dashboards, while advanced, often present challenges related to complexity and driver distraction. Analog systems remain appealing to some drivers for their simplicity, though they scored the lowest in usability (82.17).

To meet the needs of modern drivers, future dashboard designs should: Focus on intuitive digital systems with streamlined layouts, reserve familiar tactile controls in hybrid designs to bridge the gap between traditional and modern systems, and incorporate customizable features that let drivers tailor interfaces to their preferences.

Future research should evaluate these systems in real-world driving scenarios, with an emphasis on diverse demographics. Adaptive technologies that simplify interfaces during high-demand situations may further enhance safety and usability. By prioritizing these elements, automotive manufacturers can develop dashboards that seamlessly integrate advanced features with intuitive design, supporting safer and more efficient driving experiences.

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